KASHMIR COLLEGE OF ENGINEERING & TECHNOLOGY

Approved by AICTE New Delhi | Affiliated with the University of Kashmir
Campus: Ranbirgarh Narbal, Srinagar - 190017, Jammu & Kashmir | Web: https://kcet.edu.in

B.Tech in Artificial Intelligence Engineering - Complete Syllabus

Approved AICTE 4-Year Scheme & Detailed Syllabus Booklet | Category: Syllabus
Curriculum Model: AICTE Model Curriculum / University of Kashmir Ordinances

Semester 3

* AI-301: Discrete Mathematics & Graph Theory (4 Credits)
* AI-302: Data Structures & Algorithms with C++ (4 Credits)
* AI-303: Digital Electronics & Logic Design (3 Credits)
* AI-304: Python Programming for Data Science & AI (3 Credits)
* AI-305: Computer Organization & Architecture (3 Credits)
* AI-306P: Data Structures & Algorithms Lab (2 Credits)
* AI-307P: Python AI Development Lab (2 Credits)

Semester 4

* AI-401: Linear Algebra & Matrix Methods for Machine Learning (4 Credits)
* AI-402: Database Management Systems & NoSQL (4 Credits)
* AI-403: Operating Systems & Systems Programming (3 Credits)
* AI-404: Foundations of Artificial Intelligence & State-Space Search (4 Credits)
* AI-405: Object-Oriented Software Engineering (3 Credits)
* AI-406P: Artificial Intelligence & Search Lab (2 Credits)
* AI-407P: Database Management Systems Lab (2 Credits)

Semester 5

* AI-501: Machine Learning: Supervised, Unsupervised & Ensemble Methods (4 Credits)
* AI-502: Knowledge Representation & Semantic Reasoning (3 Credits)
* AI-503: Formal Languages & Automata Theory (3 Credits)
* AI-504: Computer Networks & Distributed Protocols (3 Credits)
* AI-PE1: Professional Elective I: Optimization Techniques for AI (3 Credits)
* AI-505P: Machine Learning Lab in PyTorch & Scikit-Learn (2 Credits)
* AI-506P: Knowledge Engineering & Prolog Lab (2 Credits)
* HS-501: Management Principles & Engineering Economics (2 Credits)

Semester 6

* AI-601: Deep Learning & Neural Network Architectures (CNN, RNN, Transformers) (4 Credits)
* AI-602: Computer Vision, Image Processing & Object Detection (4 Credits)
* AI-603: Reinforcement Learning & Multi-Agent Intelligent Systems (3 Credits)
* AI-604: Cloud Computing, Big Data Infrastructure & MLOps (3 Credits)
* AI-PE2: Professional Elective II: Natural Language Processing (3 Credits)
* AI-605P: Deep Learning & Computer Vision Lab (2 Credits)
* AI-606P: Cloud Infrastructure & MLOps Deployment Lab (2 Credits)

Semester 7

* AI-701: Generative AI, Large Language Models (LLMs) & Diffusion Models (4 Credits)
* AI-702: AI Ethics, Safety, Bias & Explainable AI (XAI) (3 Credits)
* AI-PE3: Professional Elective III: Edge AI & Embedded Machine Learning (3 Credits)
* OE-701: Open Elective I: Cybersecurity & Threat Intelligence (3 Credits)
* AI-703P: Capstone Project Stage-I & Research Proposal (4 Credits)
* AI-704I: Industrial Summer Internship Evaluation & Seminar (2 Credits)

Semester 8

* AI-801: Major Capstone Project Stage-II (Working Model & Defense) (10 Credits)
* AI-PE4: Professional Elective IV: Autonomous Robotics & Self-Driving Systems (3 Credits)
* OE-801: Open Elective II: Intellectual Property Rights & Tech Entrepreneurship (3 Credits)
* AI-802V: Comprehensive Technical Viva-Voce (2 Credits)